Telco Customer Churn Analysis

1. Project Overview

Customer churn is an important business problem for telecommunications companies because losing existing customers can reduce recurring revenue and increase the cost of acquiring new customers.

This project analyses customer churn for a fictional telecommunications company using a dataset containing **7,043 customers**. The objective was to identify customer groups associated with higher churn, understand the reasons customers leave, and identify potential areas where the business could improve customer retention.

The project follows an end-to-end data analytics workflow using **PostgreSQL, SQL and Power BI**.

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2. Business Problem

The telecommunications company wants to better understand why customers are leaving and which customer segments have the highest churn risk.

The analysis aims to answer the following questions:

- What percentage of customers have churned?
- Which contract types have the highest churn?
- Does customer tenure relate to churn?
- Which internet service types have the highest churn?
- Does payment method relate to customer churn?
- Are higher monthly charges associated with churn?
- Which age groups have higher churn?
- Does premium technical support relate to customer retention?
- Are some customer offers associated with higher churn?
- Why are customers leaving the company?
- Are high-value customers being lost?
- Which areas should the business prioritise for retention?

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3. Dataset

The dataset contains **7,043 customer records** and includes demographic, service, contract, billing, revenue and churn information.

Important variables used in the analysis include:

- Customer ID

- Age
- Gender
- Senior Citizen
- Tenure in Months
- Contract
- Internet Service
- Internet Type
- Payment Method
- Monthly Charge
- Total Charges
- Total Revenue
- Premium Tech Support
- Online Security
- Online Backup
- Offer
- Churn Label
- Churn Score
- CLTV
- Churn Category
- Churn Reason

The dataset also contains customer location and demographic information.

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4. Tools Used

PostgreSQL / pgAdmin

Used for:

- Storing the dataset
- Creating the customer churn table
- Importing the CSV data
- Writing SQL queries
- Performing exploratory data analysis
- Creating a reusable SQL view for Power BI

SQL

Used for:

- Customer segmentation
- Churn-rate calculations
- Aggregation
- Grouping
- Filtering
- Identifying churn patterns

- Creating analytical fields

Power BI

Used for:

- Data modelling
- KPI creation
- Interactive visualisations
- Dashboard development
- Presenting business insights

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5. Analytical Approach

The project followed an exploratory and business-focused approach.

Step 1 — Data Preparation

The raw customer churn data was imported into PostgreSQL.

A SQL view called `churn_analysis` was created to provide Power BI with a clean analytical dataset.

Additional analytical fields were created, including:

- Age Group
- Tenure Group

Age groups were defined as:

- Under 30
- 30–49
- 50–64
- 65+

Tenure groups were defined as:

- 0–12 Months
- 13–24 Months
- 25–48 Months
- 49+ Months

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6. Key Performance Indicators

The analysis identified:

- **Total Customers:** 7,043
- **Churned Customers:** 1,869
- **Retained Customers:** 5,174
- **Overall Churn Rate:** 26.54%
- **Retention Rate:** 73.46%

Approximately one in four customers in the dataset has churned.

This indicates that customer churn is significant enough to warrant further investigation rather than being treated as a minor issue.

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7. Key Findings

7.1 Churn by Contract Type

Contract Type	Customers	Churn Rate
Month-to-Month	3,610	**45.84%**
One Year	1,550	**10.71%**
Two Year	1,883	**2.55%**

Month-to-month customers have the highest churn rate at **45.84%**.

Customers on one-year contracts have a substantially lower churn rate of **10.71%**, while two-year customers have the lowest rate at **2.55%**.

This indicates a strong association between contract type and churn. However, the analysis does not establish that contract type itself causes customers to leave.

7.2 Churn by Tenure

Tenure Group	Customers	Churned	Churn Rate
0–12 Months	2,186	1,037	**47.44%**
13–24 Months	1,024	294	**28.71%**
25–48 Months	1,594	325	**20.39%**

Tenure Group	Customers	Churned	Churn Rate
49+ Months	2,239	213	**9.51%**

Customers in their first 12 months have the highest churn rate at **47.44%**.

Churn decreases consistently as tenure increases, reaching only **9.51%** among customers with 49 or more months of tenure.

This suggests that the early stages of the customer relationship are particularly important for retention.

7.3 Churn by Internet Service

Customers with internet service have a churn rate of **31.83%**, compared with **7.40%** among customers without internet service.

Further analysis showed that the type of internet service makes a significant difference.

Churn by Internet Type

Internet Type	Customers	Churn Rate
Fiber Optic	3,035	**40.72%**
Cable	830	**25.66%**
DSL	1,652	**18.58%**
None	1,526	**7.40%**

Fiber Optic customers have the highest churn rate at **40.72%**.

This suggests that internet-service customers should not be treated as one homogeneous group when developing retention strategies.

7.4 Churn by Payment Method

Payment Method	Customers	Churn Rate
Mailed Check	385	**36.88%**
Bank Withdrawal	3,909	**34.00%**
Credit Card	2,749	**14.48%**

Mailed-check customers have the highest churn rate at **36.88%**, closely followed by bank-withdrawal customers at **34.00%**.

Credit-card customers have a considerably lower churn rate of **14.48%**.

This suggests an association between payment method and churn, although other customer characteristics may contribute to this relationship.

7.5 Monthly Charges and Churn

Customer Status	Customers	Average Monthly Charge	Average Total Charges
Stayed	5,174	61.27	2,550.79
Churned	1,869	**74.44**	**1,531.80**

Churned customers have a higher average monthly charge of **74.44**, compared with **61.27** among retained customers.

However, churned customers have lower average total charges.

This may be partly explained by the earlier tenure finding: customers who leave tend to have shorter relationships with the company, giving them less time to accumulate total charges.

The results suggest that higher monthly charges may be associated with churn, particularly among newer customers.

7.6 Churn by Age Group

Age Group	Customers	Churn Rate
65+	1,142	**41.68%**
50–64	1,892	**24.42%**
30–49	2,608	**24.04%**
Under 30	1,401	**21.70%**

Customers aged 65 and above have a substantially higher churn rate of **41.68%**.

The other age groups have relatively similar churn rates, ranging from approximately 21.70% to 24.42%.

This indicates that older customers may require additional investigation to understand the factors contributing to their higher churn.

7.7 Premium Technical Support

Premium Tech Support	Customers	Churn Rate
No	4,999	**31.19%**
Yes	2,044	**15.17%**

Customers without premium technical support have a churn rate of **31.19%**, compared with **15.17%** among customers who have the service.

The churn rate is therefore more than twice as high among customers without premium technical support.

This suggests that technical support and customer service may play an important role in customer retention.

7.8 Churn by Offer

Offer	Customers	Churn Rate
Offer E	805	**52.92%**
None	3,877	**27.11%**
Offer D	602	**26.74%**
Offer C	415	**22.89%**
Offer B	824	**12.26%**
Offer A	520	**6.73%**

Offer E has the highest churn rate at **52.92%**, while Offer A has the lowest at **6.73%**.

The large variation between offers suggests a strong association between offer type and churn.

However, the offer itself should not automatically be considered the cause of churn. Customers receiving different offers may have different characteristics or risk profiles.

8. Why Customers Churn

Churn Categories

Category	Customers	% of Churn
Competitor	841	**45.00%**
Attitude	314	**16.80%**
Dissatisfaction	303	**16.21%**
Price	211	**11.29%**
Other	200	**10.70%**

The **Competitor** category represents the largest reason for customer churn, accounting for **45% of all churned customers**.

This indicates that competitive pressure is a major business concern.

Specific Churn Reasons

The most common specific reasons include:

1. Competitor had better devices — **16.75%**
2. Competitor made better offer — **16.64%**
3. Attitude of support person — **11.77%**
4. Don't know — **6.96%**
5. Competitor offered more data — **6.26%**
6. Competitor offered higher download speeds — **5.35%**
7. Attitude of service provider — **5.03%**
8. Price too high — **4.17%**
9. Product dissatisfaction — **4.12%**
10. Network reliability — **3.85%**

The two largest individual reasons were competitors offering **better devices** and **better offers**.

Together, these account for approximately **33.39% of all churned customers**.

This highlights competitive positioning as an important area for the company to investigate.

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9. High-Value Customer Analysis

A high-value churn segment was created using a CLTV threshold of **5,000 or higher**.

Using this analytical definition, the dashboard identified:

> **555 high-value customers who have churned.**

This is important because losing customers with higher lifetime value could have a greater financial impact than losing lower-value customers.

The 5,000 CLTV threshold is an analytical threshold selected for this project and is not a company-defined business rule.

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10. Power BI Dashboard

The analysis was presented through a three-page Power BI dashboard.

Page 1 — Customer Churn Overview

This page provides a high-level view of the customer base.

KPIs

- Total Customers
- Churned Customers
- Churn Rate
- Retention Rate

Visuals

- Churn Rate by Contract Type
- Churn Rate by Tenure
- Churn Rate by Internet Type

Purpose

The page provides an immediate overview of churn and highlights the customer segments with the most significant differences in churn rates.

Page 2 — Churn Drivers

This page explores the customer characteristics and services associated with higher churn.

Visuals

- Churn Rate by Offer
- Churn Rate by Payment Method
- Churn Rate by Premium Tech Support
- Churn Rate by Age Group
- Top 10 Reasons for Customer Churn
- Churn by Category

Purpose

The page helps identify potential drivers of customer churn and provides insight into why customers are leaving.

Page 3 — Customer Retention & Risk

This page focuses on retention opportunities and customer value.

KPIs and Visuals

- High-Value Churned Customers
- Churned Customers by Risk Group
- Average CLTV by Churn Status
- Churned Customers by Contract and Internet Type
- Customers Lost to Competitors

Purpose

The page focuses on identifying customers and segments that could be prioritised in future retention strategies.

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11. Business Recommendations

Based on the analysis, the following areas should be investigated by the business.

1. Focus on early-tenure customers

Customers with 0–12 months of tenure have a **47.44% churn rate**.

The company should investigate stronger onboarding, early engagement and first-year retention initiatives.

2. Investigate month-to-month customers

Month-to-month customers have a **45.84% churn rate**.

The company could investigate whether longer-term contracts, loyalty incentives or improved value propositions could encourage customers to remain longer.

3. Investigate Fiber Optic churn

Fiber Optic customers have a **40.72% churn rate**.

The company should investigate service quality, pricing, network performance, speeds and customer expectations among Fiber Optic customers.

4. Review Offer E

Offer E has the highest observed churn rate at **52.92%**.

The company should investigate which customers receive this offer and whether the offer is targeted at customers who were already at high risk of leaving.

5. Improve customer support

Customers without premium technical support have a **31.19% churn rate**, compared with **15.17%** among customers with support.

Customer service and technical support should therefore be investigated as potential retention opportunities.

6. Address competitive pressure

Competitor-related reasons account for **45% of churn**.

The business should review:

- Device offerings
- Pricing and promotions
- Data allowances
- Internet speeds
- Competitor positioning

7. Prioritise high-value customers

The analysis identified **555 high-value churned customers** based on the project's CLTV threshold.

A future retention strategy could prioritise high-value customers who also demonstrate high churn risk.

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12. Limitations

Several limitations should be considered when interpreting the analysis.

Association does not imply causation

The analysis identifies relationships between customer characteristics and churn. It does not prove that a particular factor directly causes customers to leave.

Dataset is based on a fictional telecommunications company

The findings should therefore be treated as an analytical case study rather than actual business performance.

Churn score is an existing dataset variable

The project uses the churn score provided in the dataset rather than developing an independent predictive machine-learning model.

Analytical thresholds

The CLTV threshold of 5,000 and churn-risk thresholds used in the dashboard were created for analytical segmentation and are not official company-defined thresholds.

Further statistical analysis could be performed

Future work could use statistical testing or predictive modelling to determine which variables are most strongly associated with churn after controlling for other factors.

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13. Future Improvements

Potential future extensions include:

- Building a customer churn prediction model using Python
- Testing different machine-learning algorithms
- Identifying the most important predictors of churn
- Developing a customer-level retention scoring system
- Creating automated Power BI refreshes
- Performing statistical significance testing
- Analysing customer segments using clustering
- Developing targeted retention recommendations based on customer profiles

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14. Conclusion

This project demonstrates an end-to-end approach to customer churn analysis using **PostgreSQL, SQL and Power BI**.

The analysis found that churn is particularly high among **early-tenure customers, month-to-month customers, Fiber Optic customers, customers without premium technical support, and customers associated with certain offers**.

The analysis also showed that **competitor-related factors are the largest broad category of churn**, with better devices and better competitor offers among the most common specific reasons for customers leaving.

Overall, the project demonstrates how raw customer data can be transformed into structured analysis, interactive dashboards and actionable business insights.

The key business takeaway is that customer retention should not be approached as a single problem. Different customer segments show substantially different churn patterns, meaning retention strategies should be **targeted according to customer characteristics, service experience, contract type, tenure and perceived competitive value**.

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15. Skills Demonstrated

Technical Skills

- SQL
- PostgreSQL
- pgAdmin
- Data cleaning and preparation
- Data aggregation
- Data transformation
- DAX
- Power BI
- Dashboard development
- KPI development
- Data visualisation

Analytical Skills

- Exploratory data analysis
- Customer segmentation
- Churn analysis
- Trend and pattern identification
- Business interpretation
- Root-cause investigation

- Data-driven recommendations

Business Skills

- Problem definition
- Translating data into business questions
- Identifying retention opportunities
- Communicating insights to stakeholders
- Distinguishing correlation/association from causation
- Developing actionable recommendations